UCLA Extension

395952: Fundamentals of Cybersecurity

COM SCI-X 420.1

Spring 2024 Section 1 4 Credits 04/01/2024 to 06/10/2024 Modified 04/02/2024

Meeting Times

This course takes place fully online and includes three optional scheduled live meetings through the Canvas Zoom feature. Course weeks run from Monday at 12:00 AM to Sunday at 11:59 PM Pacific Time (US) for 11 weeks.

Optional live meetings are offered as an opportunity for you to expand your knowledge in this field and connect with your instructor and classmates. The Zoom meetings are scheduled for the following dates and times:

- Week Two: Friday at 6:00 PM (PT)
- Week Six: Friday at 6:00 PM (PT)
- Week Eleven: Friday at 6:00 PM (PT)

Description

This course combines theoretical cybersecurity fundamentals with state-of-the-art use cases for a comprehensive introduction to the field. It is designed to benefit a cross-section of students, including those considering a career in cybersecurity; system administrators, network administrators or security auditors; professionals responsible for securing network services, devices and traffic communication in their organization; and information technologists with the passion and inner drive to protect organizations and understand security needs. Get practical knowledge and professional experience for implementing basic security services on any type of computer network. Topics covered are security roles, risk analysis, cryptography and network security. The course material is consistent with relevant portions of the CompTIA Security+ certification.

Objectives

During this course, students will:

- Compare security roles and security controls
- Explain threat actors and threat intelligence

- Perform security assessments and identify social engineering attacks and malware types
- Summarize basic cryptographic concepts and implement public key infrastructure
- Implement authentication controls
- Implement identity and account management controls
- Implement secure network designs, network security appliances, and secure network protocols
- Implement host, embedded/Internet of Things, and mobile security solutions
- Implement secure cloud solutions
- Explain data privacy and protection concepts
- Perform incident response and digital forensics
- Summarize risk management concepts and implement cybersecurity resilience
- Explain physical security

Course Prerequisites

We recommend the following skills and knowledge before starting this course:

- Use of a keyboard and mouse.
- Knowledge of the function and basic features of PC components.
- Use of Windows to create and manage files and use basic administrative features (Explorer, Control Panel and Management Consoles).
- Knowledge of basic network terminology and functions (such as OSI Model, topology, Ethernet, TCP/IP, switches and routers).
- Knowledge of TCP/IP addressing, core protocols, and troubleshooting tools.

Outcomes

By the end of this course, you should be able to:

- Identify the fundamental components of information security.
- Compare and contrast security controls using categories and functional types.
- Explain how to assess external and insider threat actor types in terms of intent and capability.
- Identify various threats to information security.
- Explain general procedures for conducting security assessments.
- Identify types of security assessments, such as vulnerability, threat hunting, and penetration testing to detect vulnerabilities.
- Summarize types of cryptographic function; hash algorithm, symmetric cipher, asymmetric cipher, and explain how they are used in hybrid encryption products to provide confidentiality, integrity, authentication, and resiliency.
- Be familiar with the tools and procedures used to issue different types of certificates and manage PKI operations.
- Identify and assess the design and use of authentication products for on-premises networks, web/cloud apps, and physical security in terms of meeting confidentiality, integrity, and availability requirements.
- Manage identity and access.

- Identify and use segmentation-based network designs and provision switching, routing, Wi-Fi, and load balancing technologies for secure network access.
- Be familiar with network appliances such as firewalls, proxies, IDS, and SIEM collectors/aggregators to implement secure network designs.
- Identify and use endpoint management solutions to apply device and application enforcement for mobile connection and other technologies.
- Implement security for hosts and software.
- Summarize virtualization and cloud computing concepts, and understand cloud security controls for compute, storage, and network functions.
- Explain the importance of data governance policies and tools to mitigate the risk of data breaches and privacy breaches, and implement security solutions for data protection.
- Identify and explain the process and procedures involved in effective incident response and implement strategies to remediate intrusion events.
- Explain the key aspects of digital forensics, including the secure acquisition and handling of evidence.
- Explain risk management, business impact analysis (BIA) and disaster recovering planning (DRP) processes and metrics.
- Identify redundancy, backup, configuration/change management, diversity, and deception to improve cybersecurity resilience.
- Ensure the continuity of business operations in the event of an incident.
- Explain the importance of physical security controls for access, surveillance, environmental protection, and secure data destruction.

Materials

The Official CompTIA Security+ Study Guide (SY0-601)

Author: James Pengelly
ISBN: 978-1-64274-328-9

Available from multiple sources online including:

- CompTIA Store (Academic) (https://academic-store.comptia.org/the-official-comptia-security-plus-student-guide-ebook/p/SEC-601-SDBK-20-C)
- CompTIA Store (https://store.comptia.org/the-official-comptia-security-plus-student-guide-ebook/p/SEC-601-SDBK-20-C)
- ERG Learning (https://www.erglearningcenter.com/The-Official-CompTIA-Student-Guide.php#!/The-Official-CompTIA-Security-Student-Guide-Exam-SY0-601-eBook/p/492062634/category=0)

Evaluation

Criteria

Graded Activities	Points / Percentage	Notes
Assignments	40%	• Ten total assignments. • Assignments are worth 40 points each.
Discussions	30%	• Eight total discussions. • Discussions are worth 37.5 points each.
Case Studies	10%	• Two total case studies. • Case studies are worth 50 points each.
Exams	20%	Midterm Exam • 20 questions worth 100 total points. Final Exam • 20 questions worth 100 total points.
Total	100%	

Breakdown

Grade	Range
A+	100% to 97%
A	< 97% to 93%
A-	< 93% to 90%
B+	< 90% to 87%
B	< 87% to 83%
B-	< 83% to 80%
C+	< 80% to 77%

Grade	Range
C	< 77% to 73%
F	< 73% to 0%

Grade	Range	Notes

Course Policies

Course Format: Online

- This course is designed to be completed entirely online via Canvas.
- The course is broken into weekly modules which run Monday at 12:00 am (PT) to Sunday at 11:59 pm (PT).
- You may log in at any time of the day; however, you are expected to prepare and participate in the course on a weekly basis.
- It is paramount that you set aside enough time to complete the required readings and course assignments each week. The assigned tasks should take about 6-10 hours per week depending on your prior experience.
- All formal written work must reflect the use of proper grammar and should be carefully edited.
- Grades will be based on the quality of completed course assignments, participation in discussions, exams, and case studies.

Online Discussions and Feedback

- Online interaction stimulates the learning process, so be prepared to respond to weekly online discussion questions regarding the concepts from the weekly lectures. The purpose is for you to share knowledge with your classmates, learn from them, and have meaningful conversations.
- Active participation in weekly online discussions is required in this course.
- A full explanation of questions is described in the discussion. If you have any concerns, contact the instructor for clarification.
- Your initial posts (responses to the discussion question) are due each week by **Sunday** at 11:59 pm (PT).
- Your initial posts should:
  - Be professional and respectful.
  - Be in the form of 1-2 paragraphs.
  - Describe what you learned from researching multiple articles on the discussion topic in your own words. Do not directly copy/paste from articles.
  - Provide links to where you found your information at the bottom of your post.
  - Incorporate your opinions, pertinent information from things that you've read, and examples from your experience.

    - Feature good writing, correct spelling, and mechanics.
- Responses to classmates are highly encourage and should:
    - Agree and disagree with others as long as they apply to the topic and are respectful.
    - Include more than phrases such as "I agree with that" or "Interesting comment."
    - Share your thoughts, ideas, and opinions, and expand on what your classmate posted.
    - If a classmate's post already has multiple replies, consider replying to another classmate's post to increase the overall communication in this discussion.
- All discussions will be reviewed and scored according to the assignment rubrics within 14 days of the due date.

Everyone is entitled to their opinion, but please be respectful and thoughtful with your responses to each other.

Assignments and Feedback

- Assignments are designed to assess your understanding of the key concepts presented during the week.
- All instructions are provided in the actual assignment. If you have any concerns, contact the instructor for clarification.
- All assignments are due by **Sunday** at 11:59 pm (PT) the week they are assigned.
- All assignments will be reviewed and scored according to the assignment rubrics within 14 days of the due date.

Case Study Analysis and Feedback

There are two case studies included in this course for you to analyze and apply your knowledge to realistic situations.

- All instructions are provided in the actual assignment. If you have any concerns, contact the instructor for clarification.
- Case studies are due by **Sunday** at 11:59 pm (PT) the week they are assigned.
- Case studies will be reviewed and scored according to the assignment rubrics within 14 days of the due date.

Exams

The Midterm and Final exams are comprehensive based on the information covered within the course. Each exam includes 20 multiple choice questions for a total of 100 points.

Midterm Exam

- The midterm exam will be available beginning on Friday of Week 6.
- Once started, you must complete the midterm exam within 90 minutes.
- You are given one attempt to pass the exam.

Final Exam

- The final exam will be available beginning on Friday of Week 11.
- Once started, you must complete the final exam within 3 hours.
- You are given two attempts to pass the exam. The highest score of all attempts will be recorded.

Deadlines and Late Work

- Please submit work by the due dates identified in the Schedule section of the Course Syllabus.
- Due dates for all assignments are final unless prior arrangements have been agreed to by the instructor. Points will be deducted from your grade if arrangements were not made prior to the due date.
- Life happens, and I understand that work and family emergencies occur that may prohibit you from submitting work on time. **Please notify me IN ADVANCE of the deadline** if an assignment will be submitted late.
- **No assignment will be accepted past the last day of the course.**

Planning Your Study Time

This course is 11 weeks. To plan your study time, it is estimated that you will spend about 9 hours per week reading, researching, engaging in course discussion activities, and completing assignments. Depending on the extent of your academic preparation and experience with the topics, the amount of study time needed may vary considerably.

🏛 Institutional Policies

Student Conduct

Students are subject to disciplinary action for several types of misconduct or attempted misconduct, including but not limited to academic dishonesty, such as cheating, multiple submission, plagiarism, or knowingly furnishing false information to the University; or behavioral misconduct, such as theft or misuse of the intellectual property of others, harassment, or disruption of the learning environment.

All published and unpublished material, whether in manuscript, printed or electronic form, is covered under this definition and includes the use of material generated wholly or in part through the use of artificial intelligence (except when the use of AI has received prior authorization for assessment as a reasonable accommodation for a student's disability, or when the use of AI is a specified part of the coursework, e.g. data science or user experience). Students are encouraged to familiarize themselves with the Student Rights & Responsibilities Policy (https://www.uclaextension.edu/student-services/rights-and-responsibilities) and to report concerns.

Services for Students with Disabilities

In accordance with the Americans with Disabilities Act of 1990, UCLA Extension provides appropriate accommodations and support services to qualified applicants and students with disabilities. These include, but are not limited to, auxiliary aids/services such as sign language interpreters, assistive listening devices for hearing-impaired individuals, extended time for and proctoring of exams, and registration assistance. Accommodations and types of support services vary and are specifically

designed to meet the disability-related needs of each student based on current, verifiable medical documentation. Arrangements for auxiliary aids/services are available only through UCLA Extension's Service for Students with Disabilities Office at (310) 825-7851 or by email at ODS@uclaextension.edu. For complete information, please visit Accessibility and Disability Services (https://www.uclaextension.edu/enrollment-and-support/accessibility-and-disability-services).

Incompletes

The interim grade Incomplete (I) may be approved for a student who has completed the majority of the course requirements, with passing quality (grade C or higher), but is unable to complete a small portion of the coursework by the course end date for good cause. For courses in which an Incomplete may be allowed, approval by the instructor of record and the academic program director is required. The Incomplete grade is not an option for courses that do not bear credit, such as 700, 800, or 900-level courses.

- It is the student's responsibility to petition for an Incomplete by emailing the appropriate academic program department (https://www.uclaextension.edu/contact-ucla-extension) at least one week before the end of the course. The Program Department will initiate the petition process once the email is received.
- The student, the instructor, the CE/Program Director, and the program staff must complete the petition prior to the final course meeting or before the quarter end date (https://www.uclaextension.edu/calendar). This process can take up to one week to complete.
- The instructor will approve or deny the request. The instructor will provide details on what the student needs to accomplish in order to complete the course, as well as a due date for submitting completed work. The due date cannot exceed the end of the ensuing quarter (https://www.uclaextension.edu/calendar) when a final grade must be reported or the Incomplete lapses to the grade "F," "NP," or "U." Visit UCLA Extension Grading Scale (https://www.uclaextension.edu/transcripts-credits-grades/grading-scale) for more information.

An Incomplete allows the student to complete only work that is outstanding and does not allow prior completed work to be retaken or resubmitted.

All Grades are Final

No change of grade may be made by anyone other than the instructor, and then, only to correct clerical errors. No term grade except Incomplete may be revised by re-examination. The correction of a clerical error may be authorized only by the instructor of record communicating directly with personnel of Student and Alumni Services.

Sexual Harassment

The University of California is committed to creating and maintaining a community where all individuals who participate in University programs and activities can work and learn together in an atmosphere free of harassment, exploitation, or intimidation. Every member of the community should be aware that the University prohibits sexual harassment and sexual violence, and that such behavior

violates both law and University policy. The University will respond promptly and effectively to reports of sexual harassment and sexual violence, and will take appropriate action to prevent, to correct, and when necessary, to discipline behavior that violates our policy.

All Extension students and instructors who believe they have been sexually harassed are encouraged to contact the Department of Student and Alumni Services for complaint resolution: UCLA Extension, 1145 Gayley Ave., Los Angeles, CA 90024; Voice/TTY: (310) 825-7031. For more information, please view the University's full Policy on Sexual Harassment and Sexual Violence (https://policy.ucop.edu/doc/4000385/SVSH).

Additional Items

Protecting Privacy and Data During Live Instruction

Live meeting sessions for this class, when applicable, are being conducted over Zoom. As the host, the instructor may be recording live sessions. Only the host has the ability to record meetings, no recording by other means is permitted. Recorded sessions will be posted in the Videos area of this class unless otherwise notified. Due to privacy, recordings are not available for download and are only accessible via Canvas for the duration of the class. If you have privacy concerns and do not wish to appear in the recording, do not turn on your video and/or audio. If you also prefer to use a pseudonym instead of your name, please let the instructor know what name you will be using so that the instructor knows who you are during the session. To rename yourself during a Zoom meeting, click on Participants, click on your name, click on More, click on Rename. If you would like to ask a question, you may do so privately through the Zoom chat by addressing your chat question to the instructor only (and not to ""everyone""). Additionally, chat may be used and moderated for live questions, and saving of chats is enabled. If you have questions or concerns about this, please contact the instructor via Canvas Inbox.

Pursuant to the terms of the agreement between Zoom and UCLA Extension, the data is used solely for this purpose and Zoom is prohibited from re-disclosing this information. UCLA Extension also does not use the data for any other purpose. Recordings will be deleted when no longer necessary. However, recordings may become part of an administrative disciplinary record if misconduct occurs during a video conference.

Course and Instructor Evaluation

UCLA Extension values your feedback on course and instructor evaluations. We ask all students to take a few minutes to complete an end-of-course evaluation survey. Updates to the course and instruction are influenced by your feedback. Understanding your student experience is essential to ensure continuing excellence in the online classroom and is appreciated by your instructor and the UCLA Extension academic leadership.

Your participation in a survey is voluntary, and your responses are confidential. After instructors submit grades, they will be given an evaluation report, but this report will not contain your name.

About Your Online Course Materials

Please note the following about online course components at UCLA Extension:

- Students must have basic computer skills, including the use of word processing software, email, and the ability to use internet browsers, such as Safari, Firefox, or Chrome.
- Students are responsible for meeting the technical requirements of Canvas and familiarizing themselves with the Canvas Learning Management System.
    - What are the browser and computer requirements for Canvas? (https://community.canvaslms.com/t5/Canvas-Basics-Guide/What-are-the-browser-and-computer-requirements-for-Instructure/ta-p/66)
- Students are responsible for keeping a copy of all assignments and work submitted, and to be aware of all assignments, due dates, and course guidelines.
- Students have access to courses via Canvas for an additional 30 calendar days after the course end date listed in the syllabus (the first 14 days are full access; the rest are read-only).
- Students are encouraged to download/print content throughout the duration of the course and before the additional 30-day access ends. No further access is possible after the course becomes unavailable.

  To download all your assignment submissions in Canvas, please refer to the online support guide. (https://community.canvaslms.com/docs/DOC-10606) for more information or contact Canvas Support via the help menu within Canvas.

UCLA Extension Canvas and Learning Support

For immediate 24/7 Canvas technical support, including holidays, click on **Help** (located on the menu to the left) where you can call or chat live with a Canvas Support representative.

UCLA Extension Academic Technology and Learning Innovation
The UCLA Extension Learning Support staff assists both students and instructors with Canvas-related technical support, as well as general and administrative questions.

Learning Support staff is available Monday through Friday, from 8 AM to 9 PM (Pacific Time), except holidays:

- Email: atli@uclaextension.edu (mailto:atli@uclaextension.edu)

Campus Safety Escorts

For students taking classes held on the UCLA campus and in and around Westwood Village, the UCLA Police Department provides a free walking escort service every day of the year from dusk until 1 a.m. Community Service Officers (CSOs) are available to walk students, faculty, staff members and visitors to and from anywhere on campus, in Westwood Village, and in the village apartments. CSOs are uniformed students who have received special training and are employed by the UCLA Police Department. To obtain an escort, please call (310) 794-9255 and allow 15 to 20 minutes for your escort to arrive. For complete information, see UCLA Evening Escorts (https://police.ucla.edu/cso/evening-escorts).

Schedule

When	Module Title	Notes
Week One 4/1/24 - 4/7/24	Information Security Roles, Controls, and Risks Associated to Threat Actors and Intelligence	Lesson Topics: • Comparing Security Roles and Controls • Explaining Threat Actors and Threat Intelligence Activities: • **Read lessons 1 and 2** from *CompTIA Security+* • **Read and view all information provided in the Week One Module** • **Complete Week One Discussion: Introduce Yourself** Due Sunday by 11:59 PM (PT) • **Complete Week One Assignment: Security Controls** Due Sunday by 11:59 PM (PT) • **Complete Case Study 1: Keylogger** Due Sunday by 11:59 PM (PT)
Week Two 4/8/24 - 4/14/24 Optional Zoom Meeting	Security Assessment, Social Engineering, and Malicious Tools	Lesson Topics: • Performing Security Assessment • Identifying Social Engineering and Malware Activities: • **Read lessons 3 and 4 from** *CompTIA Security+* • **Read and view all information provided in the Week Two Module** • **Complete Week Two Discussion: Social Engineering Techniques** Due Sunday by 11:59 PM (PT) • **Complete Week Two Assignment: Security Assessment** Due Sunday by 11:59 PM (PT) • **Attend Optional Zoom Meeting** Friday at 7:00 PM (PT)
Week Three 4/15/24 - 4/21/24	Cryptography and PKI	Lesson Topics: • Summarizing Basic Cryptographic Concepts • Implementing Public Key Infrastructure (PKI) Activities: • **Read lessons 5 and 6 from** *CompTIA Security+* • **Read and view all information provided in the Week Three Module** • **Complete Week Three Discussion: Cryptography** Due Sunday by 11:59 PM (PT) • **Complete Week Three Assignment: PKI** Due Sunday by 11:59 PM (PT)

When	Module Title	Notes
Week Four 4/22/24 - 4/28/24	Identity Access Management (IAM)	Lesson Topics: • Implement Authentication Controls • Implement Identity and Account Management Controls Activities: • Read **lessons 7 and 8** from *CompTIA Security+* • **Read and view all information provided in the Week Four Module** • **Complete Week Four Discussion: IAM Solutions** Due Sunday by 11:59 PM (PT) • **Complete Week Four Assignment: IAM Processes** Due Sunday by 11:59 PM (PT)
Week Five 4/29/24 - 5/5/24	Network Architecture, Design, and Security	Lesson Topics: • Implementing Secure Network Designs • Implementing Network Security Appliances • Implementing Secure Network Protocols Activities: • **Read lessons 9, 10, and 11 from *CompTIA Security+*** • **Read and view all information provided in the Week Five Module** • **Complete Week Five Discussion: Secure Organization WIFI Network** Due Sunday by 11:59 PM (PT) • **Complete Week Five Assignment: Network Architecture, Design, and Security** Due Sunday by 11:59 PM (PT)

When	Module Title	Notes
Week Six 5/6/24 - 5/12/24 Optional Zoom Meeting	Endpoint Security and Solutions	Lesson Topics: • Implementing Host Security • Implementing Secure Mobile Solutions Activities: • **Read lessons 12 and 13 from *CompTIA Security+*** • **Read and view all information provided in the Week Six Module** • **Complete Week Six Assignment: Endpoint Security** Due Sunday by 11:59 PM (PT) • **Complete Midterm Exam** Due Sunday by 11:59 PM (PT) • **Complete Mid-Course Survey** Due Sunday by 11:59 PM (PT)
Week Seven 5/13/24 - 5/19/24	Application and Cloud Security Solutions	Lesson Topics: • Summarizing Secure Application Concepts • Implement Secure Cloud Solutions Activities: • **Read lessons 14 and 15 from *CompTIA Security+*** • **Read and view all information provided in the Week Seven Module** • **Complete Week Seven Discussion: OWASP** Due Sunday by 11:59 PM (PT) • **Complete Week Seven Assignment: Cloud Security Solutions** Due Sunday by 11:59 PM (PT)
Week Eight 5/20/24 - 5/26/24	Data Privacy and Protection	Lesson Topics: • Explaining Data Privacy and Protection Concepts Activities: • **Read lesson 16 from *CompTIA Security+*** • **Read and view all information provided in the Week Eight Module** • **Complete Week Eight Discussion: Privacy and Data Sensitivity Concepts** Due Sunday by 11:59 PM (PT) • **Complete Week Eight Assignment: Privacy and Data Protection Controls** Due Sunday by 11:59 PM (PT)

When	Module Title	Notes
Week Nine 5/27/24 - 6/2/24	Data Forensic and Incident Response (DFIR)	Lesson Topics: • Performing Incident Response • Explain Digital Forensics Activities: • Read lessons 17 and 18 from *CompTIA Security+* • Read and view all information provided in the Week Nine Module • Complete Week Nine Discussion: Post Breach Attack Simulation Due Sunday by 11:59 PM (PT) • Complete Week Nine Assignment: Incident Response - Forensic Tools Due Sunday by 11:59 PM (PT)
Week Ten 6/3/24 - 3/9/24 Optional Zoom Meeting	Risk Management, Business Continuity, and Disaster Recovery	Lesson Topics: • Summarizing Risk Management Concepts • Implementing Cybersecurity Resilience Activities: • Read lessons 19 and 20 from *CompTIA Security+* • Read and view all information provided in the Week Ten Module • Complete Week Ten Discussion: Cybersecurity Resilience Due Sunday by 11:59 PM (PT) • Complete Week Ten Assignment: Risk Management Processes and Concepts Due Sunday by 11:59 PM (PT) • Attend Optional Zoom Meeting Friday at 6:00 PM (PT)

When	Module Title	Notes
Week Eleven 6/10/24 - 6/16/24	Physical Security	Lesson Topics: • Explaining Physical Security Activities: • Read lesson 21 from *CompTIA Security+* ◦ Explain the Importance of Physical Site Security Controls ◦ Explain the Importance of Physical Host Security Controls • Read and view all information provided in the Week Eleven Module • Complete Case Study 2: Globex Corporation Security Scenario Due Sunday by 11:59 PM (PT) • Complete Final Exam Due Sunday by 11:59 PM (PT)